

YANG SUN

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EDUCATION

University of Arizona **Tucson, USA**
Ph.D. in Astronomy and Astrophysics *2021 - 2026*
Supervisor: Prof. George Rieke
Thesis – *Co-Evolution of SMBH and Galaxies From Cosmic Noon to the Present Day*

M.S. in Astronomy and Astrophysics *August 2024*

Beijing Normal University **Beijing, China**
B.S. in Astronomy *2016 - 2020*

POSITIONS

Postdoctoral Research Associate/University of Arizona *2026 - Present*

Graduate Researching Assistant/University of Arizona *2021 - 2026*

Graduate Teaching Assistant/University of Arizona *ASTR250 in Fall 2023 and 2025*

RESEARCH INTERESTS

Active Galactic Nuclei; Galaxy Evolution; SF&AGN Feedback

- supermassive black hole and galaxy co-evolution across cosmic time
- AGN-driven multiphase outflow and its impact on host galaxy
- star formation and galaxy assembly history, feedback-induced galaxy quenching

PUBLICATION OVERVIEW

See the full publication list on [SciX](#)

ORCID ID: [ID 0000-0001-6561-9443](https://orcid.org/0000-0001-6561-9443)

Total Publications: 53, with citations of 1009 (August 2026)

First Author Publications: 8 (7 refereed and 1 non-refereed)

OBSERVING PROPOSALS

MMT 2026B		PI
ALMA Cycle 13	2026.1.0683.S (7.2h)	PI
ALMA Cycle 13	2025.1.0666.S (9.3h)	PI
ALMA Cycle 12	2025.1.01045.S (7.2h)	PI
ALMA Cycle 12	2025.1.01067.S (45.6h)	co-I
JWST Cycle 5	GO 10311 (26.7h)	co-I
JWST Cycle 4	GO 7335 (14.2h)	co-I

TALKS

* for invited talk

- *MIT Monday Afternoon Talk, Boston, USA *Nov. 2025*
- *Michigan Extreme Astrophysics Group Talk, Ann Arbor, USA *Nov. 2025*
- *NOIRLab Flash Talk, Tucson, USA *Sept. 2025*
- CRISOL2025 (flash talk), Toledo, Spain *May 2025*
- JWST/NIRCam Science Team Meeting, Tucson, USA *March 2025*
- Dust and Gas throughout Cosmic Time, Hiroshima, Japan *Dec. 2024*
- JWST/MIRI Meeting, Tucson, USA *Oct. 2024*
- STScI Spring Symposium (flash talk), Baltimore, USA *Apr. 2024*
- Steward Internal Symposium, Tucson, USA *Sept. 2023*
- *Columbia University Seminar, New York, USA *July 2023*
- Galaxy Formation Workshop, Tucson, USA *Apr. 2023*
- IAU Symposium 373, Busan, South Korea *Aug. 2022*

SERVICE AND OUTREACH

- Referee for PASP and AAS Journals *2025-present*
- Co-organizer of International Scholars Task Force *2025-Present*
in Steward Observatory
- Astronomy Tutoring Offered for Majors and Minors (ATOMM) program *2025*
in Steward Observatory
- Teacher for Black Holes Camp in Flandrau Planetarium *2025*
- Co-organizer of Steward Internal Symposium *2025*
- Co-organizer of Prospective Graduate Student Visit *2024*
in Steward Observatory
- Co-producer of "Numerical Methods" Online Course *2018*
at Beijing Normal University
- Volunteer at Beijing Astronomy Planetarium *2017-2019*

HONOR

- University of Arizona Department of Astronomy Graduate Student Award *2026*
for Outstanding Scholarship
- "Jingshi" First Prize Scholarship for Excellent Academic Performance *2018-2019*
- Excellent Research Projects for Undergraduates *Sept. 2019*
- "Kuangqiao" Scholarship for Excellent Academic Performance *Nov. 2018*
- Honorable Mention for Mathematical Competition in Modeling *April 2018*
- "Jingshi" Second Prize Scholarship for Excellent Academic Performance *2016-2018*

SKILLS

Programming: Python, FORTRAN, MATLAB

Packages, Software, Tools: LATEX, CASA, TOPCAT, JWST&ALMA Proposal Tools,

BAGPIPES, pPXF, galfit, galight

Language: English, Chinese

REFERENCES

Prof. George Rieke	The University of Arizona	ghrieke@gmail.com
Prof. Marcia Rieke	The University of Arizona	mrieke@gmail.com
Prof. Ann Zabludoff	The University of Arizona	aiz@arizona.edu

PUBLICATION LIST

Refereed Paper:

As First Author:

1. **Sun, Y.**, Ji, Z., D'Eugenio, F., et al. (2026), A Census of Na D-traced neutral ISM and outflows at $0.6 < z < 4$, arXiv e-prints, arXiv:2604.18522.
2. **Sun, Y.**, Ji, Z., Rieke, G. H., et al. (2026), Extreme Neutral Outflow in a Non-active Galactic Nucleus Quiescent Galaxy at $z \sim 1.3$, ApJ, 997, 140.
3. **Sun, Y.**, Rieke, G. H., Lyu, J., et al. (2025), Evolution of the $M_* - M_{BH}$ Relation from $z \sim 6$ to the Present Epoch, ApJ, 983, 165.
4. **Sun, Y.**, Lyu, J., Rieke, G. H., et al. (2025), No Evidence for a Significant Evolution of $M_{BH} - M_*$ Relation in Massive Galaxies up to $z \sim 4$, ApJ, 978, 98.
5. **Sun, Y.**, Lee, G.-H., Zabludoff, A. I., et al. (2024), Evolution of gas flows along the starburst to post-starburst to quiescent galaxy sequence, MNRAS, 528, 5783-5803.
6. **Sun, Y.**, Yuan, H., Chen, B. (2022), Validations and Corrections of the SFD and Planck Reddening Maps Based on LAMOST and Gaia Data, ApJS, 260, 17.
7. **Sun, Y.**, Deng, D.-S., Yuan, H.-B. (2021), Precision of the Chinese Space Station Telescope (CSST) stellar radial velocities, Research in Astronomy and Astrophysics, 21, 092.

As Significant Contributor:

1. Ji, Z., **Sun, Y.**, Giavalisco, M., et al. (2026), Compact Core, Extended Reach: A Bipolar kpc-Scale Elongation in a Little Red Dot at $z \approx 5.5$, arXiv e-prints, arXiv:2606.18342.
2. Ji, Z., **Sun, Y.**, Giavalisco, M., et al. (2026), Through the Veil: Ly α Illuminates the Host Galaxies of Little Red Dots, arXiv e-prints, arXiv:2606.09970.
3. Rieke, G. H., **Sun, Y.**, Lyu, J., et al., Confirming Near- to Mid-infrared Photometrically Identified Obscured AGNs in the JWST Era, 2025, ApJ, 994, 35.
4. Zhu, Y., Bonaventura, N., **Sun, Y.**, et al., SMILES Data Release II: Probing Galaxy Evolution during Cosmic Noon and Beyond with NIRSpec Medium-Resolution Spectra, 2025, arXiv e-prints, arXiv:2508.12599.
5. Deng, D., **Sun, Y.**, Wang, T., Wang, Y., Jiang, B., Infrared Excess of a Large OB Star Sample, 2022, ApJ, 935, 175.
6. Yuan, H.-B., Deng, D.-S., **Sun, Y.**, A star-based method for precise wavelength calibration of the Chinese Space Station Telescope (CSST) slitless spectroscopic survey, 2021, RAA, 21, 074.
7. Deng, D., **Sun, Y.**, Jian, M., Jiang, B., Yuan, H., Intrinsic Color Indices of Early-type Dwarf Stars, 2020, AJ, 159, 208.

Other Co-author Publications:

1. Rinaldi, P., Rieke, G. H., Wu, Z., et al., **including Sun, Y.**, Beyond the Dot: An LRD-like Nucleus at the Heart of an IR-bright Galaxy and its Implications for High-redshift LRDs, 2026, ApJ, 1006, 205.
2. De Rossi, M. E., Rieke, G. H., Williams, C. C., et al., **including Sun, Y.**, Determining Total Infrared Luminosities from Submm Measurements of High Redshift Galaxies, 2026, arXiv e-prints, arXiv:2607.28875.
3. Zhu, Y., Rieke, M. J., Carreira, C., et al., **including Sun, Y.**, PhotoIFU: NIRCcam as a Photometric Integral Field Unit for Mapping Feedback in Galaxies, 2026, arXiv e-prints, arXiv:2607.23020.
4. Ma, Z., Zhu, Y., Ji, Z., et al., **including Sun, Y.**, A quasar hatching from a buried red phase at $z = 3.7$, 2026, arXiv e-prints, arXiv:2607.22966.
5. Scholtz, J., Carniani, S., Parlanti, E., et al., **including Sun, Y.**, JADES Data Release 4 Paper II. Data reduction, analysis, and emission-line fluxes of the complete spectroscopic sample, 2026, MNRAS, 549, stag939.
6. D'Eugenio, F., Nelson, E. J., Eisenstein, D. J., et al., **including Sun, Y.**, JADES Dark Horse: demonstrating high-multiplex observations with JWST/NIRSpec dense-shutter spectroscopy in the JADES Origins Field, 2026, MNRAS, 549, stag824.
7. Duan, Q., Tacchella, S., Johnson, B. D., et al., **including Sun, Y.**, JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: stellar population catalogue for galaxies in GOODS-N and GOODS-S, 2026, arXiv e-prints, arXiv:2605.21599.
8. Rinaldi, P., Hainline, K., D'Eugenio, F., et al., **including Sun, Y.**, The Way We Tally Becomes the Tale: the Impact of Selection Strategies on the Inferred Evolution of Little Red Dots Across Cosmic Time, 2026, arXiv e-prints, arXiv:2604.07138.
9. Alberts, S., Rieke, G. H., Shivaee, I., et al., **including Sun, Y.**, Calibrating Photometric Mid-Infrared Star Formation Rates for JWST, 2026, arXiv e-prints, arXiv:2604.01326.
10. Zhu, Y., Rieke, M. J., Ho, L. C., et al., **including Sun, Y.**, A Potential Link between Nuclear Winds and Cold Gas Outflows on Kiloparsec Scales in Reionization-era Quasars, 2026, ApJ, 1000, 312.
11. Iani, E., Rinaldi, P., Torralba, A., et al., **including Sun, Y.**, JWST Reveals Two Overmassive Black Hole Candidates in Dwarf Galaxies at $z \approx 0.7$: Pushing Black Hole Searches into the Dwarf-Galaxy Regime, 2026, arXiv e-prints, arXiv:2603.17967.
12. Shields, T., Rieke, M., Hainline, K., et al., **including Sun, Y.**, JADES: low surface brightness galaxies at $0.4 < z < 0.8$ in GOODS-S, 2026, MNRAS, 546, stag202.
13. Zhang, J., Egami, E., Sun, F., et al., **including Sun, Y.**, Abundant Population of Broad H Emitters in the GOODS-N Field Revealed by CONGRESS, FRESCO, and JADES, 2026, ApJ, 997, 250.
14. Lin, X., Egami, E., Sun, F., et al., **including Sun, Y.**, The Luminosity Function and Clustering of H Emitting Galaxies at $z = 4-6$ from a Complete NIRCcam Grism Redshift Survey, 2026, ApJ, 997, 207.
15. Ma, Z., Egami, E., Zhu, Y., et al., **including Sun, Y.**, Undermassive Hosts of $z = 4-6$ AGN from JWST/NIRCcam Image Decomposition with CONGRESS, FRESCO, and JADES, 2026, arXiv e-prints, arXiv:2601.15962.
16. Sun, F., Eisenstein, D. J., D'Eugenio, F., et al., **including Sun, Y.**, JADES: Discovery of Large Reservoirs of Small Dust Grains in the Circumgalactic Medium of Massive

Galaxies at $z \sim 3.5$ through Deep JWST/NIRCam Imaging and Grism Spectroscopy, 2026, arXiv e-prints, arXiv:2601.15961.

17. Carreira, C., Robertson, B. E., Danhaive, A. L., et al., **including Sun, Y.**, JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Catalogs of inferred morphological properties of galaxies from JWST/NIRCam imaging in GOODS-N and GOODS-S, 2026, arXiv e-prints, arXiv:2601.15957.

18. Robertson, B. E., Johnson, B. D., Tacchella, S., et al., **including Sun, Y.**, JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Photometric Catalog, 2026, arXiv e-prints, arXiv:2601.15956.

19. Alberts, S., Eisenstein, D. J., Bunker, A. J., et al., **including Sun, Y.**, JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: MIRI Coordinated Parallels in GOODS-S and GOODS-N, 2026, arXiv e-prints, arXiv:2601.15955.

20. Johnson, B. D., Robertson, B. E., Eisenstein, D. J., et al., **including Sun, Y.**, JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: NIRCam Imaging in GOODS-S and GOODS-N, 2026, arXiv e-prints, arXiv:2601.15954.

21. Lin, X., Fan, X., Sun, F., et al., **including Sun, Y.**, The Large-scale Environments of Low-luminosity AGNs at $3.9 < z < 6$ and Implications for Their Host Dark Matter Halos from a Complete NIRCam Grism Redshift Survey, 2026, ApJ, 997, 61.

22. Zhu, Y., Egami, E., Fan, X., et al., **including Sun, Y.**, Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at $z = 6.3$, 2025, ApJL, 995, L5.

23. Scholtz, J., Parlanti, E., Carniani, S., et al., **including Sun, Y.**, Tentative rotation in a galaxy at $z \sim 14$ with ALMA, 2025, MNRAS, 544, L113-L120.

24. Stone, M. A., Rieke, G. H., Lyu, J., et al., **including Sun, Y.**, The $z = 7.08$ Quasar ULAS J1120+0641 May Never Reach a "Normal" Black Hole to Stellar Mass Ratio, 2025, ApJ, 993, 168.

25. Rinaldi, P., Bonaventura, N., Rieke, G. H., et al., **including Sun, Y.**, Not Just a Dot: The Complex UV Morphology and Underlying Properties of Little Red Dots, 2025, ApJ, 992, 71.

26. Woodrum, C., Shivaee, I., Witstok, J., et al., **including Sun, Y.**, JADES: The Star Formation and Dust Attenuation Properties of Galaxies at $3 < z < 7$, 2025, arXiv e-prints, arXiv:2510.00235.

27. D'Eugenio, F., Helton, J. M., Hainline, K., et al., **including Sun, Y.**, JADES and SAPPHIRES: galaxy metamorphosis amidst a huge, luminous emission-line region, 2025, MNRAS, 542, 960-981.

28. Rieke, G. H., Buiten, V. A., Goldberg, C. E., et al., **including Sun, Y.**, Low Accretion Rates in Black Holes in Late-stage Merger Ultraluminous Infrared Galaxies, 2025, ApJ, 988, 17.

29. Zhu, Y., Rieke, M. J., Ji, Z., et al., **including Sun, Y.**, A Systematic Search for Galaxies with Extended Emission Lines and Potential Outflows in JADES Medium-band Images, 2025, ApJ, 986, 162.

30. Zhu, Y., Alberts, S., Lyu, J., et al., **including Sun, Y.**, SMILES: Potentially Higher Ionizing Photon Production Efficiency in Overdense Regions, 2025, ApJ, 986, 18.

31. Hainline, K. N., Maiolino, R., Juodžbalis, I., et al., **including Sun, Y.**, An Investigation into the Selection and Colors of Little Red Dots and Active Galactic Nuclei, 2025, ApJ, 979, 138.

32. Alberts, S., Lyu, J., Shivaei, I., et al., **including Sun, Y.**, SMILES Initial Data Release: Unveiling the Obscured Universe with MIRI Multiband Imaging, 2024, ApJ, 976, 224.
33. Zhu, Y., Bakx, T. J. L. C., Ikeda, R., et al., **including Sun, Y.**, Discovery of a Unique Close Quasar DSFG Pair Linked by a [C II] Bridge at $z = 5.63$, 2024, Research Notes of the American Astronomical Society, 8, 284.
34. Alberts, S., Williams, C. C., Helton, J. M., et al., **including Sun, Y.**, To High Redshift and Low Mass: Exploring the Emergence of Quenched Galaxies and Their Environments at $3 < z < 6$ in the Ultra-deep JADES MIRI F770W Parallel, 2024, ApJ, 975, 85.
35. Shivaei, I., Alberts, S., Florian, M., et al., **including Sun, Y.**, A new census of dust and polycyclic aromatic hydrocarbons at $z = 0.72$ with JWST MIRI, 2024, A&A, 690, A89.
36. Ji, Z., Williams, C. C., Rieke, G. H., et al., **including Sun, Y.**, Extended hot dust emission around the earliest massive quiescent galaxy, 2024, arXiv e-prints, arXiv:2409.17233.
37. Wevers, T., Coughlin, E. R., Pasham, D. R., et al., **including Sun, Y.**, Live to Die Another Day: The Rebrightening of AT 2018fyk as a Repeating Partial Tidal Disruption Event, 2023, ApJL, 942, L33.

Non-Refereed Paper:

1. **Sun, Y.**, Lee, G.-H., Zabludoff, A. I., et al., Evolution of Gas Flows over the Starburst to Post-Starburst to Quiescent Galaxy Sequence, 2023, Resolving the Rise and Fall of Star Formation in Galaxies, 373, 193-198.